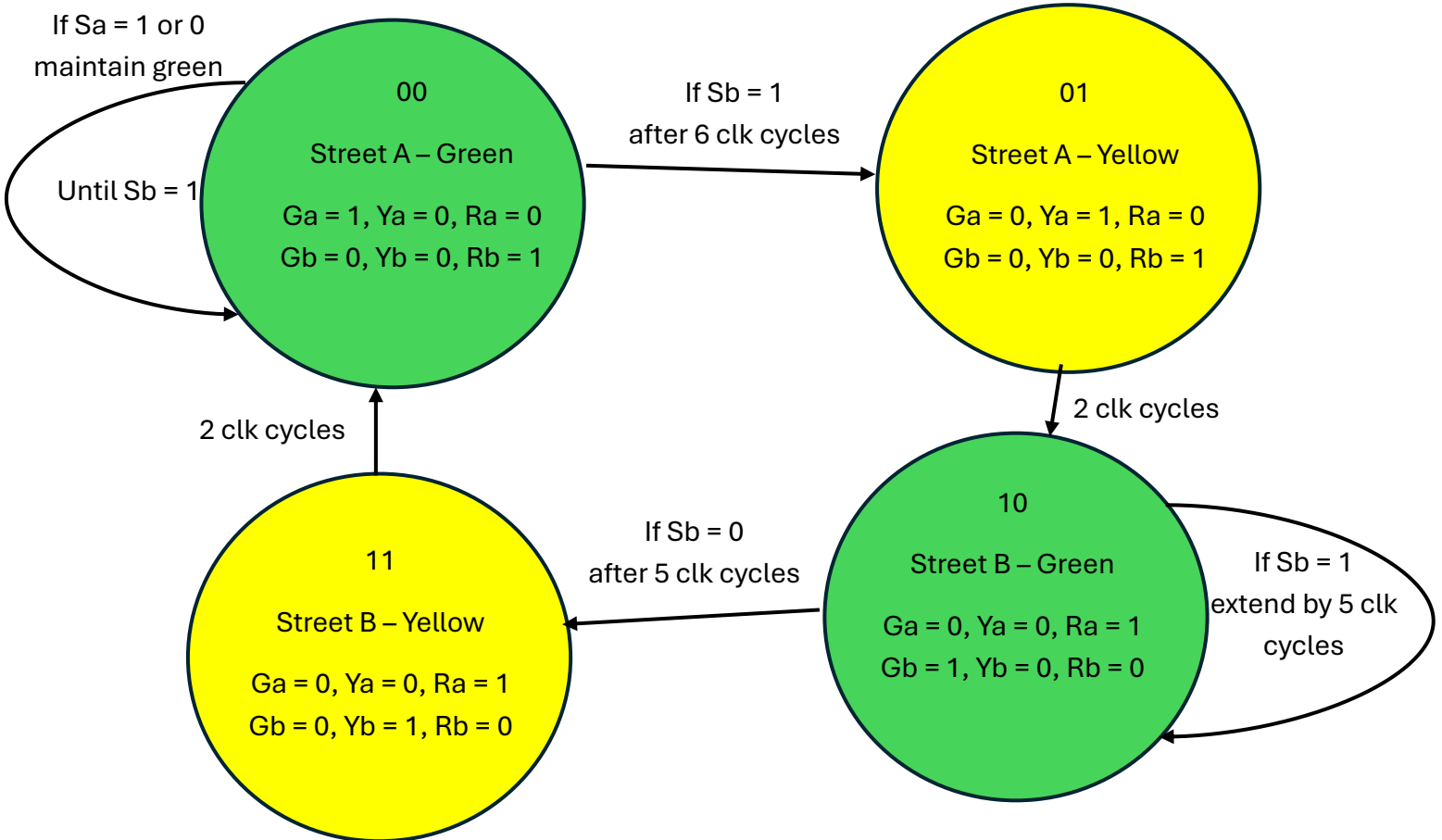


Homework 2

Problem 3:

a) Traffic Controller design as a finite state machine:



The design is safe because by using a finite state machine, the light can never have Ga and Gb on at the same time. In fact, it should never be allowed to be anything other than the 4 states 00, 01, 10, and 11 with only one being able to be on at a time. Not to mention a failsafe has been added where if for some reason both Ga and Gb are on then Ra and Rb will flash on and off every clock period.

b) Code in Folder

- c) Although a but tough to see up close I have assured the lights work as they should. At the end you can even see that the failsafe also worked as intended with the flashing Ra and Rb.

